



Ocular-motor Deception Test

2003 - present


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Others

- Pooja Patnaik
- Andrea Webb
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Agenda

- Overview of ocular sensor technologies
- Lab and field research on the ODT
- Pilot / field study discussion



Krapohl Kircher Hacker

Science Team

5

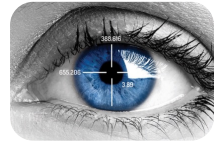
- Psychological experiments are based on theory.
- Some theories are not as well developed as in physical or biological sciences.
- Sometimes predictions based on theories are wrong.
- The variable manipulated is deception.
- Manipulation with ground truth using mock crimes (lab).
- Or, manipulation by selecting people who are highly likely to be innocent (field).

→ consensus

Science

6

- Manipulate and measure variables to see their effects on ocular-motor measures.
 - Effects of motivation
 - Presentation format
 - Presentation modality (A,V or A/V)
 - Gender
 - Semantic properties of test questions (items)
 - Language
 - Culture



→ consensus

Theoretical Assumptions

7

- Deception requires cognitive effort
 - Effects of cognitive effort
 - Pupil enlargement
 - Increased reading and re-reading
 - Decreased blink rate
- Deception testing is associated with emotional arousal
 - Effects of negative emotional reaction to a test question
 - Pupil enlargement
 - Respiration, EDA, cardiovascular activity, etc.

→ consensus

Lab Research

8

Mock Crime Experiments

9

- o Advertise in Craigslist or local websites. We do not target students.
- o Callers who meet criteria are given appointment
- o Subjects receive informed consent and instructions prior to any contact with research personnel
- o Subjects are randomly assigned to guilty and innocent conditions
- o All subjects told that some subjects steal a credit card, other subjects steal \$20, and the remaining subjects are innocent of both crimes
- o Guilty subjects steal \$20 from a secretary's purse
- o Innocent subjects do not steal anything
- o All subjects are told to deny involvement in either crime
- o All subjects are promised a cash bonus if they are able to pass the test

Test Phase

10



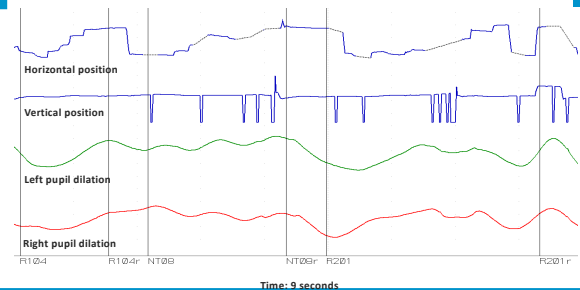
Data from Eye Tracker in 100 ms.

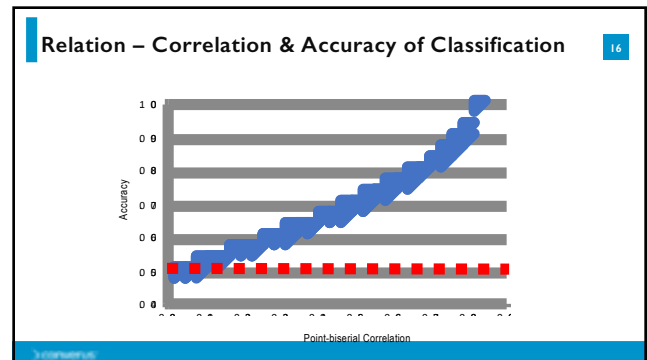
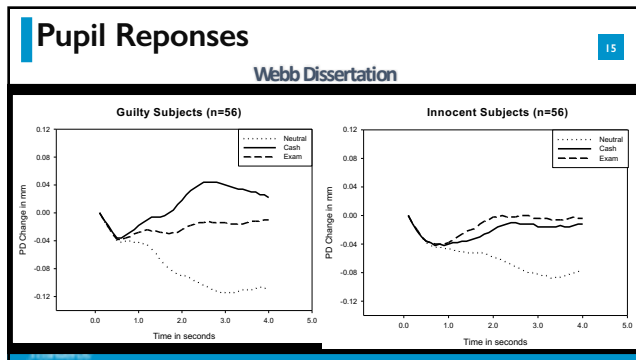
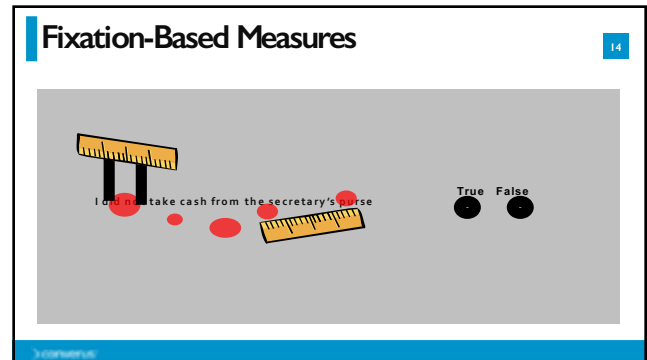
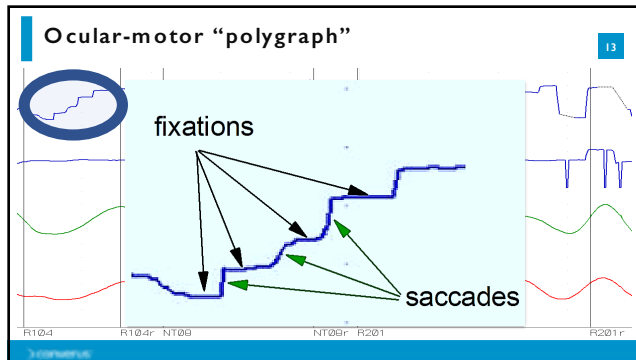
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Index	Time	HPOS	VPOS	Ldiam	Rdiam	
1	0.017	594.64	428.55	3.91	3.89	3.
2	0.033	595.08	427.40	3.90	3.89	3.
3	0.050	595.62	409.49	3.90	3.88	3.
4	0.067	595.78	363.28	3.90	3.88	3.
5	0.083	595.78	363.28	3.90	3.88	3.
6	0.100	594.30	358.06	3.91	3.88	3.
7	0.117	594.30	358.06	3.91	3.88	3.

Ocular-motor "polygraph"

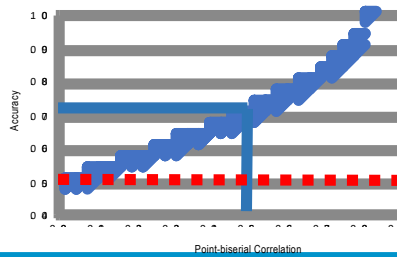
12





Relation – Correlation & Accuracy of Classification for Truthful or Deceptive Subjects

17



Feature Validity Coefficients (r_{pb})

18

	Osher (2006)	Webb (2008)	Palnalk (2015)	Palnalk et al. (2016)	Kircher & Rasidin (2016)	Mean
Sample Size	40	112	80	145	154	
Pupil Size						
Area under curve	0.550	0.464	0.586	0.546	0.484	0.517
Level of answer	NA	0.523	0.585	0.587	0.536	0.556
Reading						
Number of fixations	-0.555	-0.529	-0.406	-0.139	-0.202	-0.310
First pass duration	-0.075	-0.530	-0.253	-0.452	-0.074	-0.301
Reread duration	-0.562	-0.489	-0.342	-0.192	-0.287	-0.332
Behavioral						
Response time	-0.489	-0.480	-0.497	-0.544	-0.474	-0.499
Error rate	NA	0.057	0.093	0.056	-0.370	-0.071
Eye blink rate						
Item blink rate	NA	-0.071	-0.388	-0.260	-0.059	-0.175
Next item blink rate	NA	0.079	-0.088	0.049	0.023	0.025

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Decision Model

20

Binary logistic regression model

$$\Pr(\text{Deceptive}) = 1 / (1 + \exp(-b_0 - b_1X_1 - b_2X_2 - \dots - b_nX_n))$$

X 's are ocular-motor measures such as PD, reading RT, ER, etc.
 b 's are weights to separate truthful and deceptive groups

Decision Rule:

If $\Pr(\text{Deception}) > .50$, then "Deceptive"
 If $\Pr(\text{Deception}) < .50$, then "Truthful"

Percent Correct Decisions: Standard Lab Conditions

Experiment	Independent variables	N	nG	nI	Guilty	Innocent	Mean	ValidationG	ValidationI	Mean
Other (2006)a	Issues; serial format	40	20	20	85.0	85.0	85.0	85.0	70.0	77.5
Webb (2008)b	Sec motivation; difficulty	112	56	56	82.1	89.2	85.7	89.3	80.4	84.9
Patnaik (2013)a	Direct interrogation	48	24	24	83.3	95.8	89.6	83.3	83.3	83.3
Patnaik (2015)a	Distributed item types; pretest feedback; post-response interval	80	40	40	82.5	90.0	86.3	80.0	90.0	85.0
Patnaik et al. (2016)c	Language; culture	145	82	63	84.1	87.3	85.7	81.9	87.5	84.7
Middle Eastd	Language; culture	112	51	61	80.4	88.5	84.5			
Middle Easte	Language; culture	101	52	49				75.0	85.7	80.4
Standard Protocol		638	325	313	82.8	89.0	85.9	82.1	84.1	83.1

21

Percent Correct Decisions: Nonstandard Lab Conditions

Experiment	Independent variables	N	nG	nI	Guilty	Innocent	Mean
Other (2006)	Issues; parallel format	40	20	20	70.0	95.0	82.5
USTAR	Indirect interrogation; nonrandom assignment	71	47	27	59.6	77.8	68.7
NSA	Standardization	94	51	43	72.5	88.4	80.5
NSA	Validation	60	34	26	50.0	80.8	65.4
Patnaik 2013	Indirect interrogation	48	24	24	58.3	79.2	68.6
Patnaik (2015)	Blocked presentation	80	40	40	77.5	85.0	81.3
Nonstandard Protocol		393	216	180	65.3	84.5	74.9

22

Field Research

23

Field Validation

Objective

- To develop and validate a decision model for pre-employment screening using actual job applicants (field cases).

Ground Truth

- R1 – drug use
- R2 – religious terrorism
- Positive drug test results or admissions made independently of ODT outcomes were used as ground truth.
- We obtained confirmed truthful cases by conducting ODTs that addressed two issues with low base rates of occurrence.

24

Exclusionary Criteria

25

Criteria	Number Excluded
Confession outside scope of relevant questions	8
Applicant confessed while answering test questions	8
Applicant tested positive for benzodiazepines or hydrocodone	3
Applicant responded randomly	1

Field Validation Sites

26

- Procuraduría General de la República – Mexico 2012
 - Rogelio Lopez Maya, Claudia Soledad Pacheco Hernandez, Enrique Cabrera Aguilar (PGR), Luisa Vallejo, José Gonzalez, Utah Science Team
- Policía Federal – Mexico 2016
 - Enrique Cabrera, María Luisa Saucedo (Policía Federal); María Fernanda Gadea (Centro Mexicano de Analisis Poligrafico y Psicologico (CEMAPP))
- Instituto Nacional de Migración – Mexico 2016
 - Rogelio Lopez Maya, María Fernanda Gadea (Centro Mexicano de Analisis Poligrafico y Psicologico (CEMAPP))

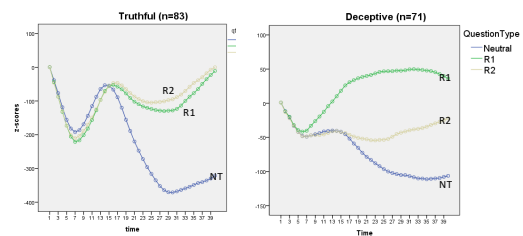
Field Data

27

Site	Issues	Approx. # Tested	# Confirmed	% Confirmed
PGR	Drugs (R1) Corruption (R2)	400	26	6
INM	Drugs (R1) Terrorism (R2)	250	30	9
FP	Drugs (R1) Terrorism (R2)	515	15	3
INM	Espionage (R1) Sabotage (R2)	83	83	100

Change in Pupil Size - 4 Seconds

28



N-fold Validation Steps

29

- Original data randomly divided into 5 equal-sized samples.
- One sample is retained as the validation data for testing the model.
- Remaining 4 subsamples are used as training data.
- Cross-validation process is repeated 5 times (the folds), with each of the 5 subsamples used exactly once as the validation data.
- The 5 results from the folds are averaged to produce a single estimation.

Advantage— all observations are used for both training and validation and each observation is used for validation exactly once.

154 confirmed cases

N-fold Validation Steps

30

- Remove the first subgroup
- Develop a decision model using the remaining four subgroups

$$\Pr(\text{Deceptive}) = 1 / (1 + \exp(b_0 + b_{X1} + b_{X2} + \dots + b_{Xk}))$$

- Use this model to classify the cases in the first subgroup
- Record the accuracy of classifications for truthful and deceptive applicants
- The observed accuracies for truthful and deceptive cases are unbiased estimates of the accuracy of the model in the target population

N-fold Validation Steps

31

- Remove the second subgroup.
- Develop a new decision model using the remaining four subgroups.

$$\Pr(\text{Deceptive}) = 1 / (1 + \exp(b_0 + b_{X1} + b_{X2} + \dots + b_{Xk}))$$

- Use this model to classify the cases in the second subgroup.
- Record the accuracy of classifications for truthful and deceptive applicants.
- The observed accuracies for truthful and deceptive cases are unbiased estimates of the accuracy of the model in the target population.
- Repeat this process for the remaining three subgroups.

5-Fold Validation

32

	Percent Correct						
Out = .50	Fold 1 N=30	Fold 2 N=30	Fold 3 N=31	Fold 4 N=31	Fold 5 N=32	Mean	Mean N=154
Truthful	100	100	76.5	94.1	94.1	92.9	87.3
Deceptive	92.9	71.4	100	57.1	86.7	81.6	

	Percent Correct						
Out = .33	Fold 1 N=30	Fold 2 N=30	Fold 3 N=31	Fold 4 N=31	Fold 5 N=32	Mean	Mean N=154
Truthful	87.5	93.8	64.7	94.1	88.2	85.7	85.1
Deceptive	92.9	71.4	100	71.4	86.7	84.5	

Conclusions

33

- ▶ The accuracy achieved in a field validation study is similar to the accuracy achieved in lab experiments.
- ▶ The accuracy achieved by the ODT is similar to that achieved with polygraph techniques.
- ▶ The accuracy of ODT on cross-validation exceeded standards for pre-employment screening established by the APA.

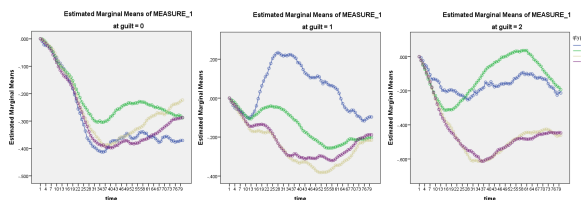
Next Steps

34

- ▶ Lab studies
 - Develop an audio version of the ODT
 - Develop other new protocols
 - Explore additional psychophysiological measures
 - Investigate effects of countermeasures
- ▶ Field studies
 - Continued efforts to identify confirmed deceptive cases will improve the accuracy and dependability of the decision model
 - Develop collaborative relationships with government and private agencies to collect field data, corroborate the present findings, and evaluate new test protocols
 - Test new protocols: (1) RRRr protocol and (2) Directed Lie Comparison Test

Pirate (RRRr) Protocol

35



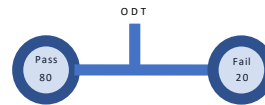
Characteristics of Field Validity Research

36

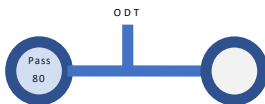
- ▶ Scientists will collaborate with the agency to create the test
- ▶ Subjects will be drawn from the population of interest
- ▶ Subjects will be tested first with the ODT, then polygraph
- ▶ The ODT will evaluate two issues covered on the polygraph tests (e.g., drugs & espionage)
 - Issues covered on the polygraph test may be more general than those on the ODT as long as admissions are specific to the issues covered on the ODT
- ▶ Ground truth
 - Deceptive cases: Confessions to the polygraph examiner before, during, or after the polygraph test
 - Truthful cases: a second test is developed with extremely low base rate issues with high face validity
- ▶ ODT scientists will be blind to ground truth
- ▶ Decisions from the ODT will be compared to ground truth by some independent party
- ▶ Depending on the results, another study may be conducted with a new decision model optimized for the population

Comparison of ODT to Polygraph

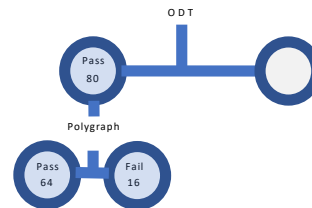
Expected Agreement on 100 Truthful Subjects with False Positive Rate = 20%

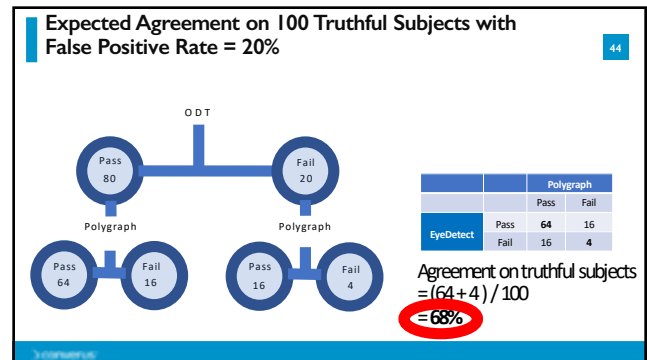
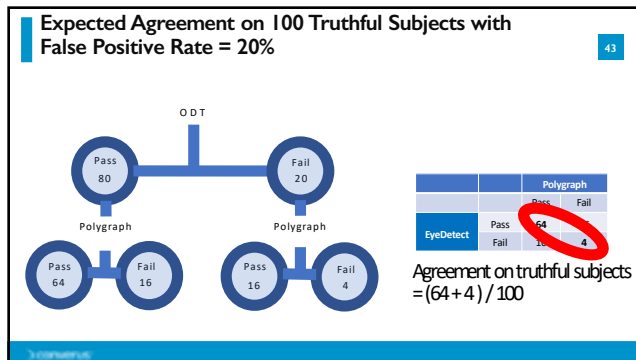
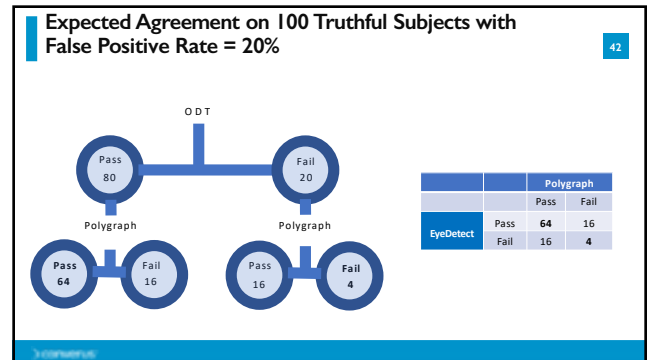
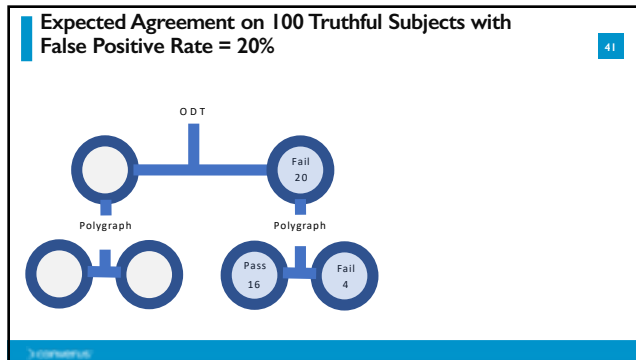


Expected Agreement on 100 Truthful Subjects with False Positive Rate = 20%



Expected Agreement on 100 Truthful Subjects with False Positive Rate = 20%





Agreement Depends on...

45

- FP and FN rate for ODT
- FP and FN rate for polygraph
- Base rate of deception
- Independence between the ODT and the polygraph
- Sample size

Dr. Kircher's Opinion

46

- Too many variables affect agreement to make sense of polygraph and ODT study results
- Unless ground truth is known, the study will not provide evidence of criterion-related validity (accuracy) or construct validity
- Life is too short to conduct research that has little if any scientific value

Publications

47

1. Cook, A.E. & Hader, D.J. (2014). The Ocular-Motor Deception Test (ODT) for detecting deception. *Journal of Applied Psychology*, 99(4), 1037-1050.
2. Hader, D.J., Kuhlman, B.B., Kircher, J.C., Cook, A.E., Woltz, D.J. (2014). Detecting deception using ocular metrics during reading. In Raskin, D.C., Honts, C.R., & Kircher, J.C. (Eds.), *Credibility Assessment: Scientific Research and Applications*. Academic Press.
3. Kircher, J.C. (2018). Ocular-motor Deception Test. In P. Rosenfeld (Ed.), *Detecting Concealed Information and Deception*. Academic Press.
4. Kircher, J.C. & Raskin, D.C. (2016). Laboratory and field research on the ocular-motor deception test. *European Polygraph* 10(4), 159-172.
5. Raskin, D.C., Hader, D.J., Kircher, J.C., Cook, A.E., Woltz, D.J., & Frander, R. (2014). Ocular-motor methods for detecting deception: Direct versus indirect interrogation. Poster presented at the Society for Psychophysiological Research convention, Atlanta, GA. (abstract)
6. Patrnick, P., Kircher, J.C., Cook, A.E., Hader, D.J., & Woltz, D.J. (2014). Ocular-motor methods for detecting deception: Direct versus indirect interrogation. Poster presented at the Society for Psychophysiological Research convention, Atlanta, GA. (abstract)
7. Patrnick, P., Woltz, D.J., Hader, D.J., Cook, A.E., Frander, R., Webb, A.K., & Kircher, J.C. (2016). Generalizability of an ocular-motor test for deception to a Mexican population. *International Journal of Applied Psychology*, 6(1), 1-9.
8. Patrnick, P., Woltz, D.J., Hader, D.J., Cook, A.E., Webb, A.K., Raskin, D.C., & Kircher, J.C. (2015). Ocular-motor detection of deception in laboratory settings. (2015, March). Paper presented at the American Psychology and Law Society convention, San Diego, CA.
9. Webb, A.K., Hader, D.J., Osher, D., Cook, A.E., Woltz, D.J., Kristjansson, S.K., & Kircher, J.C. (2009). Eye movements and pupil size reveal deception in computer administered questionnaires. In D. O. Schmonow, L.V. Estabrooke, & M. Grootjen (Eds.), *Foundations of Augmented Cognition: Neuroergonomics and Operational Neuroscience* (553-562). Berlin/Heidelberg: Springer-Verlag.
10. Webb, A.K., Honts, C.R., Kircher, J.C., Bernhardt, P.C., & Cook, A.E. (2009). Effectiveness of pupil diameter in a probable lie comparison question test for deception. *Legal and Criminal Psychology*, 14(2), 279-292.

Thank you